

Christoffel Symbols and Example on an Elliptic Paraboloid

Introduction

We proceed step by step: first the logic of Christoffel symbols, then a concrete computation on the elliptic paraboloid

$$(t, v, t^2 + v^2).$$

1 What Christoffel Symbols Represent

We consider

$$\Gamma_{ij}^k,$$

the Christoffel symbols of the second kind.

Intuition

They encode how the **basis vectors change from point to point** on a curved manifold.

In Euclidean space with Cartesian coordinates, basis vectors are constant, so

$$\Gamma_{ij}^k = 0.$$

On a curved surface (or curvilinear coordinates), the basis vectors vary, hence

$$\Gamma_{ij}^k \neq 0.$$

2 Formal Definition

Given a metric g_{ij} , the Christoffel symbols are defined by

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

where:

- g_{ij} is the metric tensor,
- g^{ij} is the inverse metric,
- $\partial_i = \frac{\partial}{\partial x^i}$.

3 Geometric Viewpoint

They describe covariant differentiation:

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^k \partial_k.$$

They:

- define a connection,
- describe parallel transport,
- appear in geodesic equations:

$$\frac{d^2 x^k}{ds^2} + \Gamma_{ij}^k \frac{dx^i}{ds} \frac{dx^j}{ds} = 0.$$

4 Elliptic Paraboloid

Consider the parametrization

$$\mathbf{X}(t, v) = (t, v, t^2 + v^2).$$

4.1 Step 1: Tangent Vectors

$$\mathbf{X}_t = (1, 0, 2t), \quad \mathbf{X}_v = (0, 1, 2v).$$

4.2 Step 2: Metric Tensor

$$g_{ij} = \langle \mathbf{X}_i, \mathbf{X}_j \rangle.$$

Compute:

$$g_{tt} = 1 + 4t^2, \quad g_{tv} = 4tv, \quad g_{vv} = 1 + 4v^2.$$

Thus:

$$g = \begin{pmatrix} 1 + 4t^2 & 4tv \\ 4tv & 1 + 4v^2 \end{pmatrix}.$$

4.3 Step 3: Inverse Metric

$$\det g = (1 + 4t^2)(1 + 4v^2) - (4tv)^2 = 1 + 4t^2 + 4v^2.$$

Let

$$D = 1 + 4t^2 + 4v^2.$$

Then:

$$g^{-1} = \frac{1}{D} \begin{pmatrix} 1 + 4v^2 & -4tv \\ -4tv & 1 + 4t^2 \end{pmatrix}.$$

4.4 Step 4: Derivatives of the Metric

$$\partial_t g_{tt} = 8t, \quad \partial_v g_{tt} = 0,$$

$$\partial_t g_{tv} = 4v, \quad \partial_v g_{tv} = 4t,$$

$$\partial_t g_{vv} = 0, \quad \partial_v g_{vv} = 8v.$$

5 Computation of Christoffel Symbols

5.1 Examples

Γ_{tt}^t

$$\Gamma_{tt}^t = \frac{4t}{D}.$$

$$\Gamma_{tv}^t = \Gamma_{vt}^t$$

$$\Gamma_{tv}^t = \frac{4v}{D}.$$

$$\Gamma_{vv}^t$$

$$\Gamma_{vv}^t = -\frac{4t}{D}.$$

5.2 Second Coordinate

$$\Gamma_{tt}^v = -\frac{4v}{D},$$

$$\Gamma_{tv}^v = \frac{4t}{D},$$

$$\Gamma_{vv}^v = \frac{4v}{D}.$$

6 Final Result

With

$$D = 1 + 4t^2 + 4v^2,$$

we obtain:

$$\Gamma_{tt}^t = \frac{4t}{D}, \quad \Gamma_{tv}^t = \frac{4v}{D}, \quad \Gamma_{vv}^t = -\frac{4t}{D}$$

$$\Gamma_{tt}^v = -\frac{4v}{D}, \quad \Gamma_{tv}^v = \frac{4t}{D}, \quad \Gamma_{vv}^v = \frac{4v}{D}.$$

7 Interpretation

- The Christoffel symbols are nonzero because the surface is curved.
- They vanish at $(0, 0)$, reflecting local flatness.
- Their structure reflects the symmetry of the paraboloid.

8 Big-Picture Intuition

Christoffel symbols can be understood as:

- correction terms to ordinary derivatives,
- objects encoding curvature indirectly,
- arising from non-straight coordinate systems.